Date: Roll No.

# III SEMESTER B.Tech (ECE, EVDT, EIOT)

## MID-SEMESTER EXAMINATION SEPTEMBER, 2026

Course Code- ECECC302 /EIECC302/ VTECC302 Course Title- Signals & Systems

Time: 1:30 Hrs Max Marks: 20

*Attempt all questions. Missing data / information (If any) may be suitably assumed & mentioned in the answer.*

| Q. No. | | Questions | Marks | CO |
|---|---|---|---|---|
| Q 1 | a | Express the following functions in terms of shifted/flipped versions of the unit step function [i.e $u(\pm t - t_0)$ *OR* $u[n-N]$],<br>i. $x(t) = u(2t+6)$<br>ii. $y(t) = u\left(-\frac{t}{4}+2\right)$<br>iii. $z[n] = u[3n-7]$<br>In each case sketch the function. | 03 | CO1 |
| | b | If $\int_{-\infty}^{\infty} y(\tau)\, x(t-\tau)\, d\tau = \sum_{k=0}^{n} x(t-kT_0)$, then y(t) is | 01 | CO1 |
| Q 2 | a | Consider a continuous-time system with input x(t) and output y(t) related by y(t) = 3x(sin t). Is the system stable, causal & time invariant | 03 | CO1 |
| | b | Compute the time period of $x[n] = e^{\frac{j2\pi n}{10}} + e^{\frac{j6\pi n}{10}} + e^{\frac{j2\pi n}{15}} + e^{\frac{j2\pi n}{30}}$ | 01 | CO1 |
| Q 3 | a | A continuous-time LTI system has an impulse response h(t) defined as:<br>$h(t) = e^{at}u(t-1) + e^{bt}u(-t+1)$<br>where a and b are real-valued constants.<br>i Determine the value(s) of a, b required for this system to be causal.<br>ii Determine the exact range of values for a and b such that the system is BIBO stable<br>iii Compute the step response of the system for a=0.5, b=2. | 03 | CO2 |
| | b | A bounded periodic signal with time period T is applied as input to an LTI system with impulse response $h(t) = e^{-|t|}$. Is the output also periodic and bounded? | 01 | CO2 |
| Q 4 | a | An LTI system's impulse response h[n] is unknown except for two facts:<br>a. it is causal and has finite duration of exactly 3 samples,<br>b. when x[n] = δ[n] + δ[n−2] is applied, y[n] = {1, 0, 3, 0, 2} for n = 0 to 4.<br>i Reconstruct h[n].<br>ii Compute the output when the input $z[n] = na^n u[n]$ | 03 | |
| | b | Determine whether the discrete-time LTI system defined by the input-output relationship $y(n) = \sum_{k=n-2}^{n+2} x[k]$ is causal. Justify your answer based on the impulse response of the system. | 01 | CO1 |
| Q 5 | a | Consider a continuous-time LTI system characterized by the impulse response $h(t) = e^{-2t}u(t)$. An input signal $x(t) = e^{j6\pi t}.\sum_{k=-\infty}^{\infty}\delta(t-k)$ is applied as input to the system<br>i Is x(t) periodic?<br>ii If Periodic, compute FS of input and output | 03 | CO1 |
| | b | A continuous-time periodic signal x(t) has a Fourier series representation with Fourier coefficients $X_k$ and a time period $T_0$=2. Determine the Fourier series representation of y(t)=x(2t-3). | 01 | CO1 |